

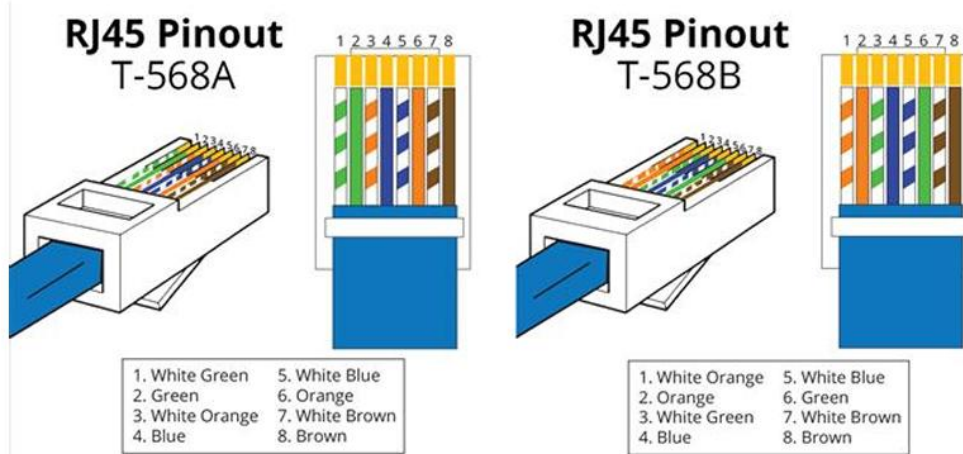


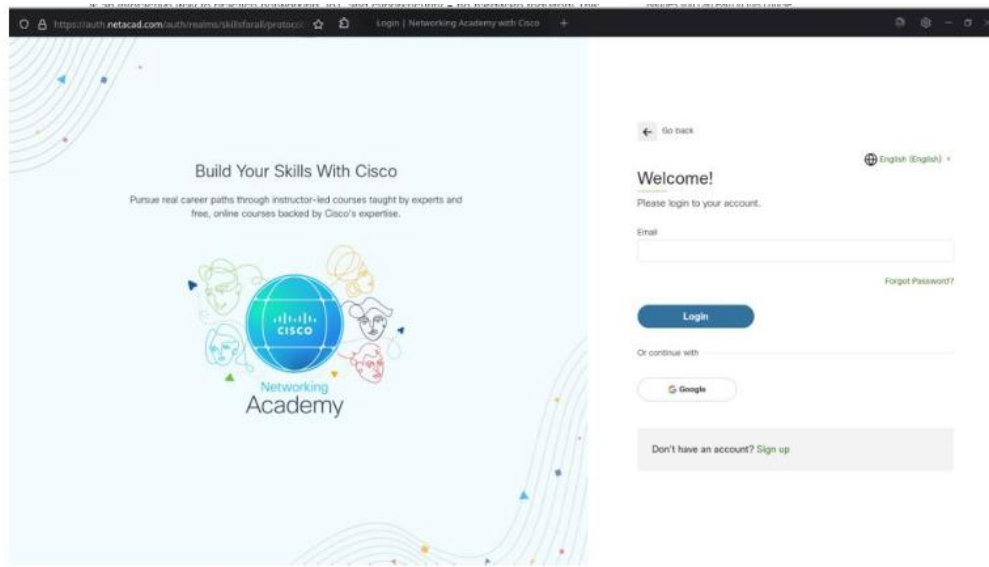
Figure 1.1: RJ45 pins arrangement



Figure 1.2: Straight Through and Cross over cable

The screenshot shows the Cisco Networking Academy website interface. The main heading is "Getting Started with Cisco Packet Tracer". Below the heading, it states "This course is part of the Learning Collections - Cisco Packet Tracer". The course is described as "Your on-ramp to Cisco Packet Tracer. Get familiar with the simulation environment and download the latest version."

The course is available in English (English) and has 2,412,198 already enrolled students. The course is self-paced and free. The curriculum is divided into two sections: Overview and Curriculum. The Overview section includes a description of the course and a list of achievements.



Learning Resources



Cisco Packet Tracer

Cisco Packet Tracer, an innovative network configuration simulation tool, helps you hone your networking configuration skills from your desktop. Use Packet Tracer to experiment while building, managing & securing infrastructures.

To obtain and install your copy of Cisco Packet Tracer, please follow these simple steps:

Step 1. Download the version of Packet Tracer you require.

- [Packet Tracer 8.2.2 MacOS 64bit](#)
- [Packet Tracer 8.2.2 Ubuntu 64bit](#)
- [Packet Tracer 8.2.2 Windows 64bit](#)

Step 2. Launch the Packet Tracer install program.

Step 3. Launch Cisco Packet Tracer by selecting the appropriate icon.

Step 4. When prompted, click on Skills For All green button to authenticate.

Step 5. Cisco Packet Tracer will launch and you are ready to explore its features.

If you need more guidance, please follow the [Cisco Packet Tracer Download and Installation Instructions](#).

System Requirements:

Computer with either Windows (10, 11), MacOS (10.14 or newer) or Ubuntu (20.04, 22.04) LTS operating system, amd64(x86-64) CPU, 4 GB of free RAM, 1.4 GB of free disk space



Download the Virtual Machine file and follow the setup instructions from the course.

- [CyberSecurity Essentials Virtual Machine for Intel or AMD CPUs](#)
- [CyberSecurity Essentials Virtual Machine for ARM CPUs \(Apple M1/M2\)](#)



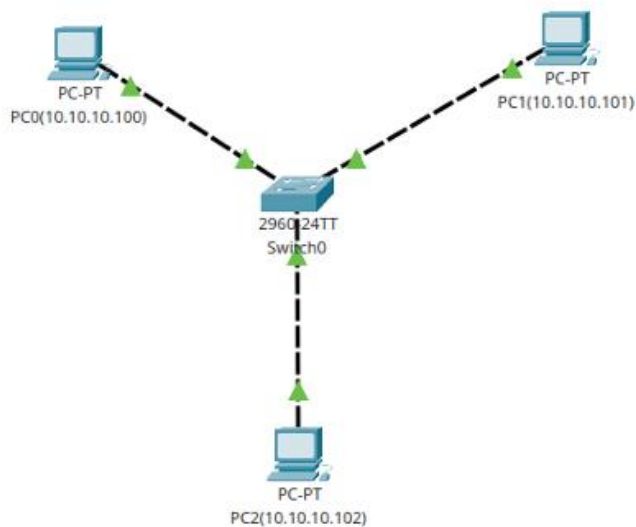
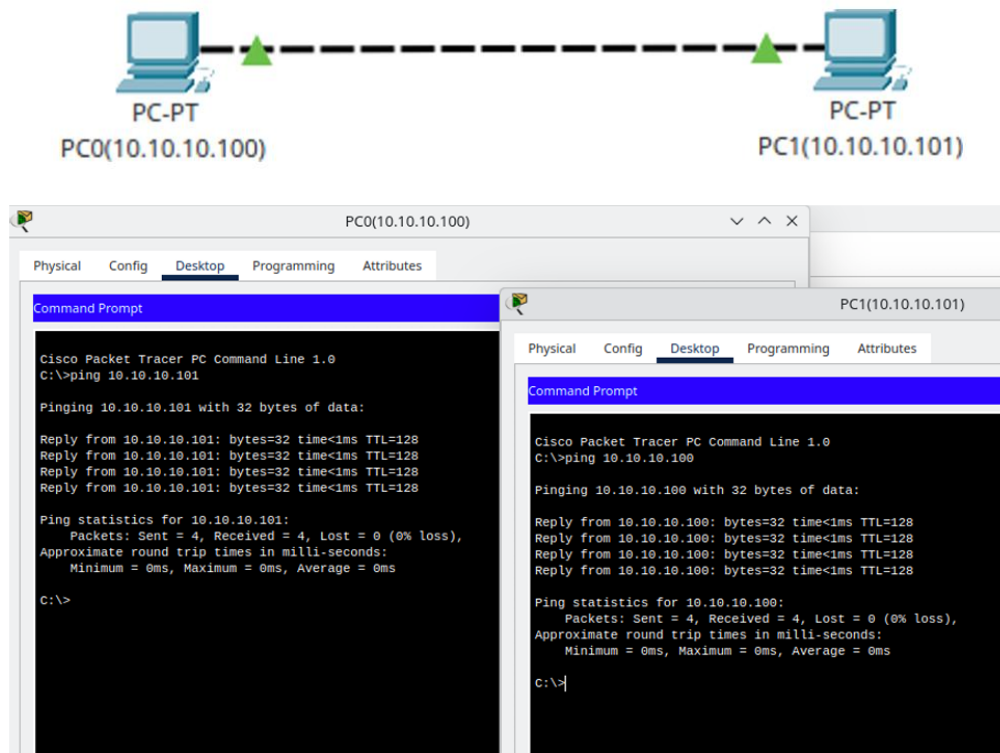


Figure 4.1: LAN using Switch

PC0(10.10.10.100)

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.101

Pinging 10.10.10.101 with 32 bytes of data:

Reply from 10.10.10.101: bytes=32 time<1ms TTL=128
Reply from 10.10.10.101: bytes=32 time<1ms TTL=128
Reply from 10.10.10.101: bytes=32 time<1ms TTL=128
Reply from 10.10.10.101: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.101:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.102

Pinging 10.10.10.102 with 32 bytes of data:

Reply from 10.10.10.102: bytes=32 time<1ms TTL=128
Reply from 10.10.10.102: bytes=32 time<1ms TTL=128
Reply from 10.10.10.102: bytes=32 time<1ms TTL=128
Reply from 10.10.10.102: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.102:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

PC2(10.10.10.102)

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.100

Pinging 10.10.10.100 with 32 bytes of data:

Reply from 10.10.10.100: bytes=32 time<1ms TTL=128
Reply from 10.10.10.100: bytes=32 time<1ms TTL=128
Reply from 10.10.10.100: bytes=32 time<1ms TTL=128
Reply from 10.10.10.100: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.101

Pinging 10.10.10.101 with 32 bytes of data:

Reply from 10.10.10.101: bytes=32 time<1ms TTL=128
Reply from 10.10.10.101: bytes=32 time<1ms TTL=128
Reply from 10.10.10.101: bytes=32 time<1ms TTL=128
Reply from 10.10.10.101: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.101:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

PC1(10.10.10.101)

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.100

Pinging 10.10.10.100 with 32 bytes of data:

Reply from 10.10.10.100: bytes=32 time<1ms TTL=128
Reply from 10.10.10.100: bytes=32 time<1ms TTL=128
Reply from 10.10.10.100: bytes=32 time<1ms TTL=128
Reply from 10.10.10.100: bytes=32 time<1ms TTL=128|

Ping statistics for 10.10.10.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.102

Pinging 10.10.10.102 with 32 bytes of data:

Reply from 10.10.10.102: bytes=32 time<1ms TTL=128
Reply from 10.10.10.102: bytes=32 time<1ms TTL=128
Reply from 10.10.10.102: bytes=32 time<1ms TTL=128
Reply from 10.10.10.102: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.102:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

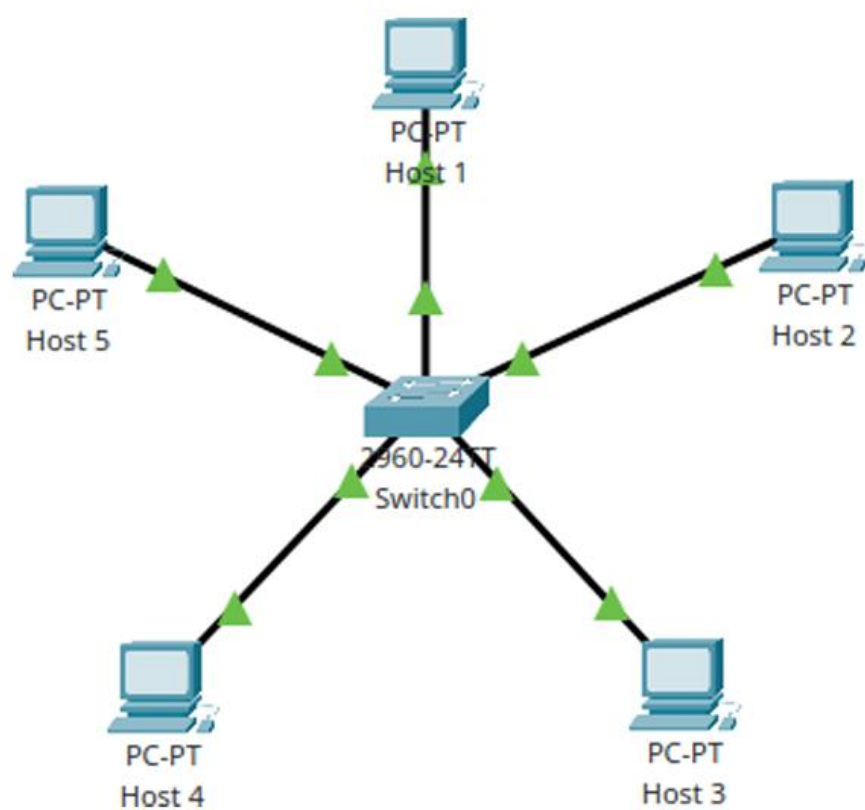


Figure 5.1: Star Topology

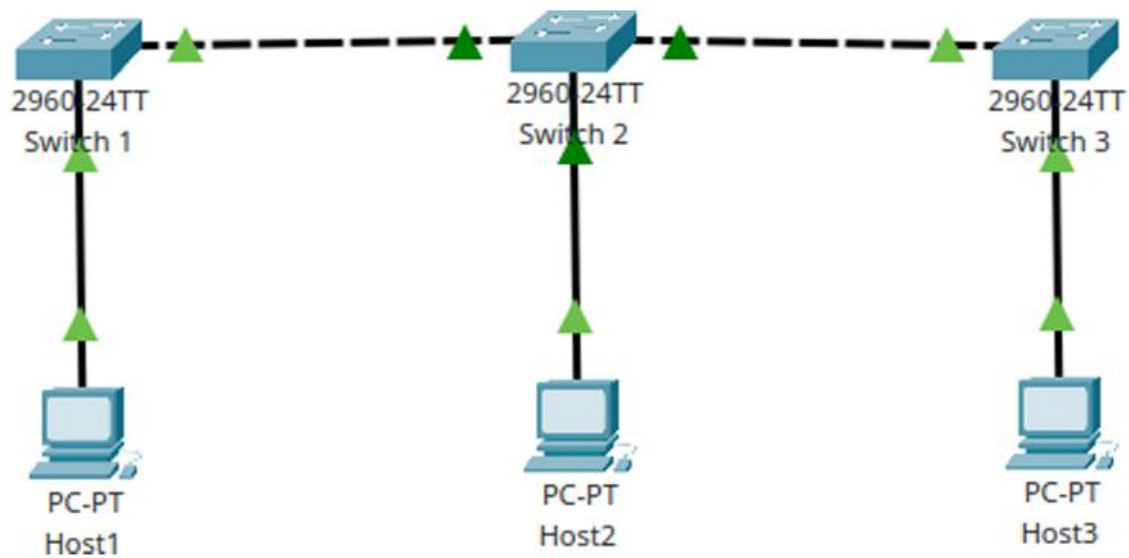


Figure 5.2: Bus Topology

Host 1

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 192.168.1.103: bytes=32 time<1ms TTL=128
Reply from 192.168.1.103: bytes=32 time<1ms TTL=128
Reply from 192.168.1.103: bytes=32 time<1ms TTL=128
Reply from 192.168.1.103: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.104

Pinging 192.168.1.104 with 32 bytes of data:

Reply from 192.168.1.104: bytes=32 time<1ms TTL=128
Reply from 192.168.1.104: bytes=32 time<1ms TTL=128
Reply from 192.168.1.104: bytes=32 time=2ms TTL=128
Reply from 192.168.1.104: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.104:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.1.105

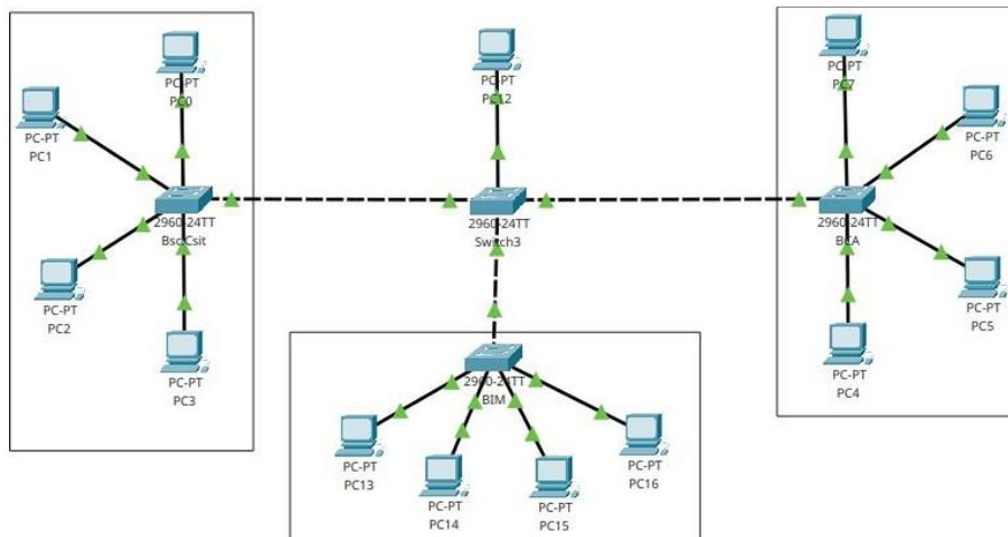
Pinging 192.168.1.105 with 32 bytes of data:

Reply from 192.168.1.105: bytes=32 time=4ms TTL=128
Reply from 192.168.1.105: bytes=32 time<1ms TTL=128
Reply from 192.168.1.105: bytes=32 time<1ms TTL=128
Reply from 192.168.1.105: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.105:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>|
```

Figure 5.3: Ping Test



PC12

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 192.168.1.114: bytes=32 time<1ms TTL=128
Reply from 192.168.1.114: bytes=32 time<1ms TTL=128
Reply from 192.168.1.114: bytes=32 time<1ms TTL=128
Reply from 192.168.1.114: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.114:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.1.115

Pinging 192.168.1.115 with 32 bytes of data:

Reply from 192.168.1.115: bytes=32 time=1ms TTL=128
Reply from 192.168.1.115: bytes=32 time=4ms TTL=128
Reply from 192.168.1.115: bytes=32 time<1ms TTL=128
Reply from 192.168.1.115: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.115:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.1.116

Pinging 192.168.1.116 with 32 bytes of data:

Reply from 192.168.1.116: bytes=32 time<1ms TTL=128
Reply from 192.168.1.116: bytes=32 time<1ms TTL=128
Reply from 192.168.1.116: bytes=32 time<1ms TTL=128
Reply from 192.168.1.116: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.116:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

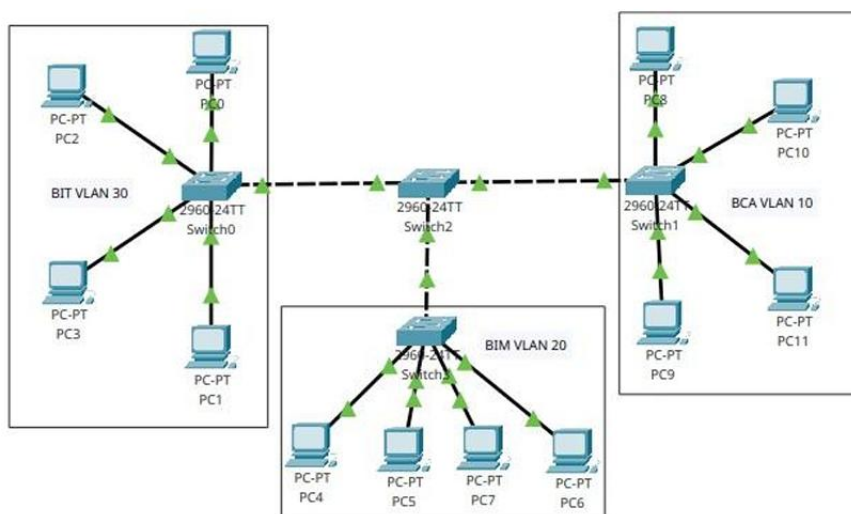


Figure 7.1: Access LAN Network

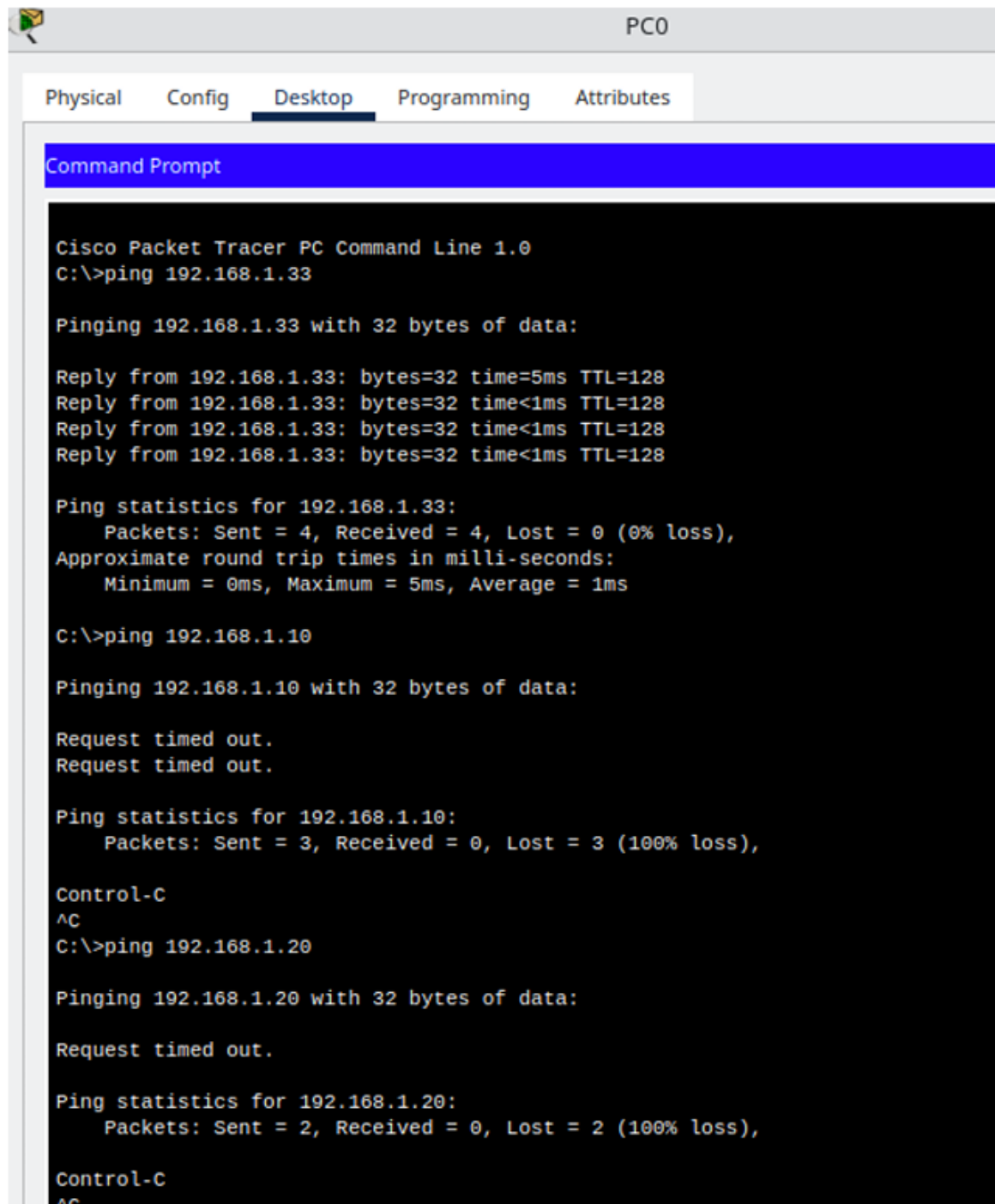
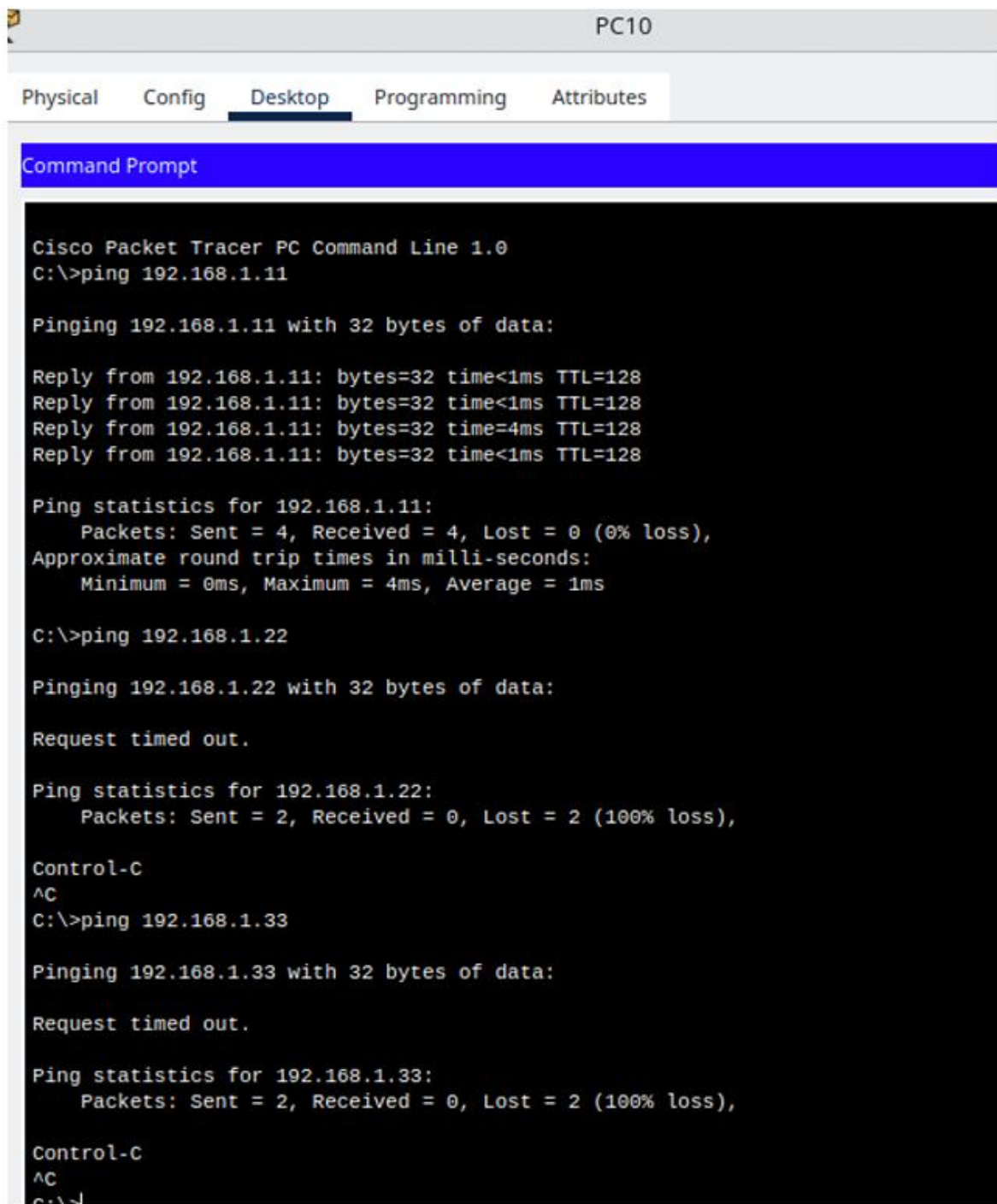


Figure 7.2: Ping test from PC0



The screenshot shows a Cisco Packet Tracer interface with the 'Desktop' tab selected for PC10. A Command Prompt window is open, displaying the results of three ping tests. The first test to 192.168.1.11 is successful. The second test to 192.168.1.22 and the third test to 192.168.1.33 both fail with 100% packet loss. The user has entered 'Control-C' twice to interrupt the tests.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.1.22

Pinging 192.168.1.22 with 32 bytes of data:

Request timed out.

Ping statistics for 192.168.1.22:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),

Control-C
^C
C:\>ping 192.168.1.33

Pinging 192.168.1.33 with 32 bytes of data:

Request timed out.

Ping statistics for 192.168.1.33:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),

Control-C
^C
C:\>
```

Figure 7.3: Ping test from PC10

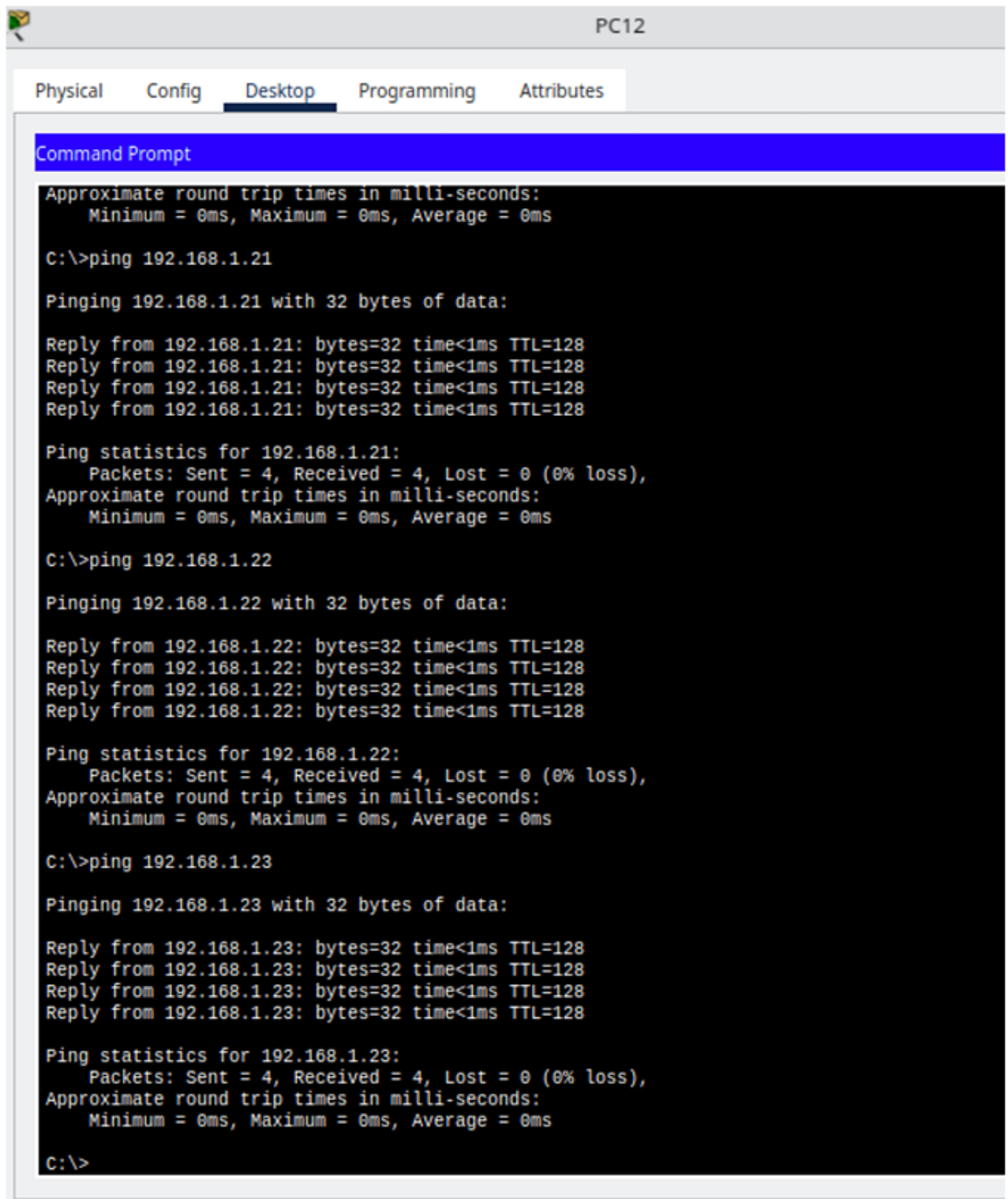


Figure 8.2: Ping Test

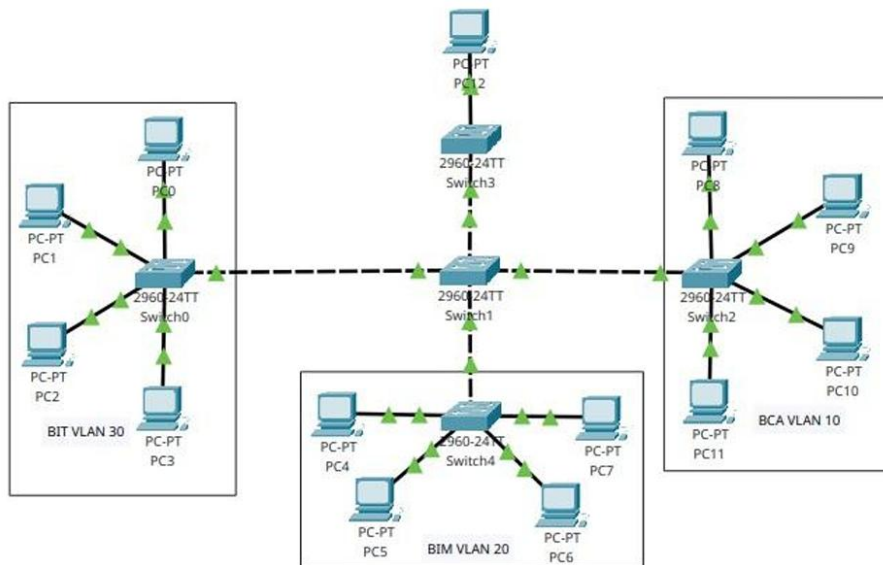
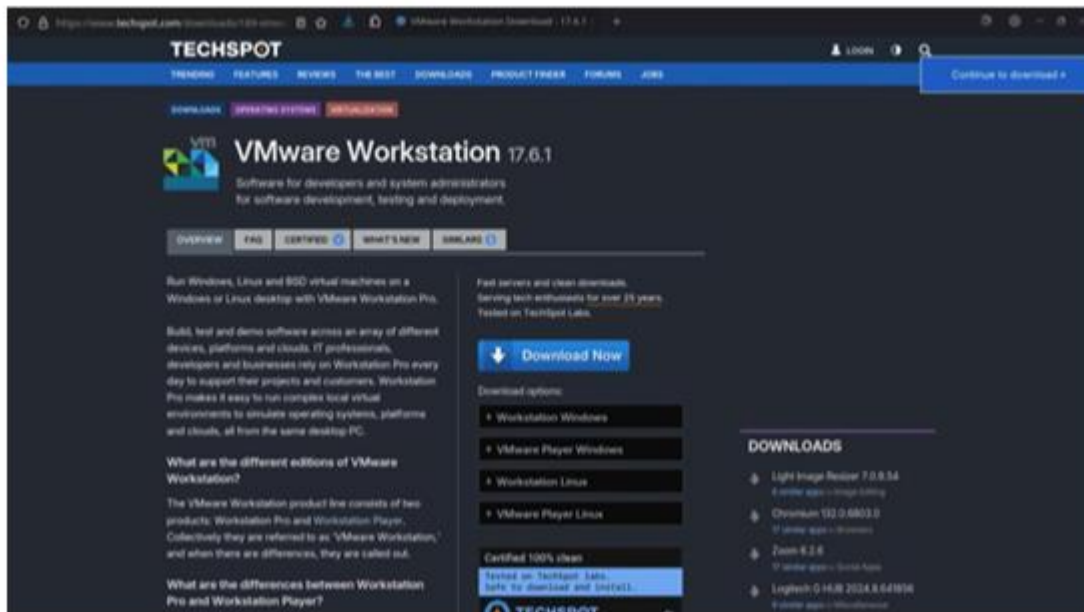


Figure 8.1: VLAN Trunking



```

~/Downloads$ sudo ./VMware-Workstation-Full-17.6.1-24319823.x86_64.bundle
[sudo] password for simpleshun:
Extracting VMware Installer...done.
Installing VMware Workstation 17.6.1
  Configuring...
[#####] 100%
Installation was successful.
simpleshun@kidman ~/Downloads$

```

